

Brandon Brown

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EDUCATION

Eastern Michigan University | GPA 3.98

Ypsilanti, MI

Bachelor of Science: Computer Science

May 2028

Awards: Presidential Scholarship(full-ride academic scholarship), Rising Star Award, Dean's List

Relevant Coursework: Data Structures & Algorithms, Intro to Web Programming, Intro to Object-Oriented Programming

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java

Frameworks/Libraries: React.js, Next.js, Express.js, Node.js, FastAPI

Tools/Technologies: Jira, Supabase, GitHub/Git, MongoDB Compass

Databases: MongoDB, PostgreSQL

PROFESSIONAL EXPERIENCE

Rocket | Incoming *Software Engineering Intern* | Detroit, MI

May 2026-Aug 2026

Jetzy | *Full-Stack Developer intern* | Remote

Sep 2025-Dec 2025

- Developed responsive, accessible web pages for **4,000+** users using **React**, **Next.js**, and **TailwindCSS**
- Built **12+ production-ready** components and full-stack features, translating Figma designs into user-friendly interfaces
- Optimized MongoDB queries with pagination and indexing, improving database efficiency and reducing page load times by ~15%
- Implemented OAuth-based authentication with Next.js server-side routes, enabling secure login and protected API endpoints

AccelerateKID | *Lead Tech Ed Instructor Intern* | Novi, MI

Mar 2025-Sep 2025

- Instructed **100+ students** (grades 4–9) in Python, JavaScript, and Java, achieving **99% project completion rate**
- Adapted quickly to new technologies and delivered structured lessons across multiple programming domains
- Provided real-time debugging support, code reviews, and technical feedback to improve student outcomes

PROJECTS

Gig Driver Peak Time Optimizer (Hackathon Winner) | *Next.js, FastAPI, MongoDB, Base44 AI*

- Built a full-stack analytics system that processes earnings data to identify peak revenue windows and generate AI-driven scheduling recommendations to increase gig drivers revenue
- Engineered a **FastAPI** analytics service using pre-aggregation to achieve **<200ms** response time for real-time insights
- Built a real-time analytics dashboard in **Next.js + TypeScript** to surface earnings trends and AI-driven recommendations
- Developed an AI-powered recommendation system (**Base44**) that processes multi-dimensional earnings data to generate actionable scheduling guidance

AI-Generated Coding Quiz Platform | *React, TypeScript, FastAPI, PostgreSQL, OpenAI*

- Built a full-stack web application that generates AI-powered coding challenges using OpenAI APIs
- Built a type-safe **React + TypeScript** frontend and a high-performance FastAPI backend with **REST APIs**, keeping non-AI processing under **15ms**
- Achieved sub-millisecond endpoint latency and **~5ms** database query time through optimized **SQL** queries and indexing
- Designed system to separate AI inference from application logic, with **~97%** of latency attributed to model inference (~2.05s total)

Spotify Playlist Generator | *React, Node.js, JavaScript*

- Developed a web app that lets users generate and save custom Spotify playlists, reducing Spotify playlist creation time by **20%**
- Integrated **OAuth 2.0** and the Spotify Web API to securely authenticate users and dynamically create, fetch, and manage playlists
- Built a responsive, component-based frontend using React hooks to support real-time user interactions
- Implemented production-grade logging and performance monitoring, achieving **0–2 ms** server-side latency for cached route

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Management Leadership for Tomorrow | *Career Prep Fellow*

Nov 2025-Present

- Selected for a competitive 18-month program preparing high-achieving students for top-tier tech and business roles
- Engaged with partners, including Capital One, Apple, and Google, on technical and career development

National Society of Black Engineers | *Vice President* | Ypsilanti, MI

Apr 2025-Present

- Analyzed engagement data to improve outreach strategies and increase participation in technical events to **100+ attendees**
- Organized professional development programs and industry-facing events for underrepresented students